

Javier Fernandez-Budiman

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EDUCATION

University of Washington, Seattle | Information School

Seattle, Washington

B.S in Informatics & Data Science

Expected Graduation June 2028

- **Relevant Coursework:** Programming with Data, Advanced Programming with Data, Probability & Statistics, Discrete Math, Fundamentals of Computer Science, Linear Algebra, Calculus I-IV

EXPERIENCE

Machine Learning Research Assistant

Seattle, Washington

UW Najafian Laboratory

February 2026-Present

- Developed an end-to-end **PyTorch** training pipeline using **AFFMAE** for Glomerular Basement Membrane image segmentation, accelerating training convergence by **10%**
- Implemented a **Python** script utilizing morphological skeletonization to quantify average pixel width from segmentation masks, automating kidney disease severity assessment.
- Automated data preprocessing and experiment tracking in **Python** cutting setup time by **2 hours** and standardizing reproducibility for research workflows
- Utilized **PowerBI** and **Matplotlib** to visualize performance evaluation on different sets of training strategies

Project Lead

Seattle, Washington

UW AI collective

May 2026-June 2026

- Led a team to build a predictive analytics platform scoring 651 Seattle intersections, processing 150,000+ historical crash records utilizing **Streamlit**
- Developed a **Negative Binomial regression model** trained on collision data, isolating the impact of key features to predict crash frequencies.
- Implemented **Empirical-Bayes shrinkage** to correct for variance in raw crash data, stratifying high risk locations into 5 severity tiers.

PROJECTS

Wildlife Jaguar Re-Identification

- Built a deep learning pipeline with 5-fold ensemble inference using **ConvNeXt + ArcFace** to re-identify individual jaguars from 1,895 camera trap images across **31 jaguars**.
- Engineered data validation using **perceptual hashing** to cluster 1,074 burst duplicates and StratifiedGroupKFold to eliminate cross-fold leakage
- Applied **k-Reciprocal Re-ranking**, improving baseline mAP by **9%**

Retail Sales Forecasting

- Built an **sklearn ensemble** pipeline (XGBoost, LightGBM, LinearSVR) predicting sales across 33 products and 54 stores, achieving **0.38 RMSLE** on a 16-day horizon
- Engineered 54 features (lag variables, rolling averages, oil price signals, holiday/payday indicators) from 6 external datasets
- Prevented data leakage with **16-day lag offsets** and nested cross-validation (TimeSeriesSplit + RandomizedSearchCV)

Leadership

Camp Manager

Sattle, Washington

Pedalheads

June 2025-August 2025

- Led operations for Pedalheads' second-largest site (100 + daily campers)

SKILLS

Languages & Tools : Python, SQL, Git, Jupyter, Docker, AWS

Data, ML & Vision : PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, SciPy, Statsmodels, XGBoost, LightGBM, YOLO, timm, Tableau, Power BI, Matplotlib, Seaborn